

Credit Risk Scorecard: Model Development Report

Give Me Some Credit Dataset

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1 Executive Summary

This report documents the end-to-end development of a points-based credit risk scorecard built on the *Give Me Some Credit* dataset. The scorecard uses Weight of Evidence (WoE) binning and logistic regression to assign a numeric score to each borrower — a higher score indicates a lower probability of serious delinquency within the next two years.

The model achieves an **AUC of 0.853**, a **Gini coefficient of 0.706**, and a **KS statistic of 54.8 percentage points** on a held-out test set of 45,001 accounts. These figures place the scorecard comfortably in the *strong* discriminatory range by standard retail-credit benchmarks (Gini > 0.50). The scorecard is calibrated to a base score of 600 with a PDO (Points to Double the Odds) of 20, making it straightforward to set and communicate application cut-offs to operational teams.

2 Data Overview

2.1 Dataset

The dataset is the publicly available *Give Me Some Credit* training file from Kaggle, originally released to support a credit default prediction competition. It contains 150,000 consumer credit observations, each representing one borrower, with 10 predictor variables and one binary target variable.

2.2 Variable Descriptions

Table 1: Predictor variables after snake_case renaming

Variable (model name)	Description	Type
serious_dlq_in_2yrs	Target: borrower was 90+ days past due in 2 yrs (1 = bad)	Binary
revolving_utilization	Total balance on revolving accounts divided by credit limit	Continuous
age	Age of borrower in years	Integer
n_times_30_59_dpd	Number of times 30–59 days past due in last 2 years	Count
debt_ratio	Monthly debt payments divided by gross monthly income	Continuous
monthly_income	Gross monthly income in USD	Continuous
n_open_credit_lines	Number of open credit lines and loans	Count
n_times_90_days_late	Number of times 90+ days past due in last 2 years	Count
n_real_estate_loans	Number of mortgage and real-estate loans	Count
n_times_60_89_dpd	Number of times 60–89 days past due in last 2 years	Count
n_dependents	Number of dependents in family (excl. borrower)	Count

2.3 Missing Values

Two variables contained missing values. Medians were imputed using training-set values only to prevent data leakage.

Table 2: Missing value counts before imputation

Variable	Missing (train)	Missing (test)	% of total
monthly_income	20,838	8,893	19.8%
n_dependents	2,670	1,254	2.6%

`monthly_income` has the highest missingness at nearly 20% of the full dataset. This is large enough to warrant caution: if income is missing non-randomly (e.g., self-employed or informal workers are systematically absent), median imputation may introduce bias. This is noted as a limitation in Section 6.

2.4 Class Imbalance

The target variable is highly imbalanced: **6.68% of borrowers** (10,026 of 150,000) are classified as bad. A stratified 70/30 split preserved this ratio in both sets:

Table 3: Class distribution after stratified split

Split	Good	Bad	Total
Train	97,981 (93.32%)	7,018 (6.68%)	104,999
Test	41,993 (93.32%)	3,008 (6.68%)	45,001

3 Methodology

3.1 Weight of Evidence Binning

Weight of Evidence (WoE) is a monotonic transformation that converts each raw predictor into a single numeric value expressing how strongly a given range of that predictor predicts good vs. bad outcomes. For each bin b of a variable, WoE is defined as:

$$\text{WoE}_b = \ln\left(\frac{\text{Distribution of Goods}_b}{\text{Distribution of Bads}_b}\right)$$

A positive WoE means the bin contains proportionally more goods than bads (lower risk); a negative WoE signals the opposite. The Information Value (IV) of a variable summarises its total predictive power by summing the WoE-weighted differences across all bins:

$$\text{IV} = \sum_b (\text{Dist. Goods}_b - \text{Dist. Bads}_b) \times \text{WoE}_b$$

WoE binning is standard in credit scorecard development because it handles non-linear relationships automatically, makes the model monotonically interpretable, and allows mixed variable types (continuous and count) to be treated uniformly. Bins are derived exclusively from the training set and then applied to the test set, preventing any leakage of test-set information into the feature engineering step.

3.2 Logistic Regression for Scorecards

Once all predictors are expressed as WoE values, logistic regression models the log-odds of default as a linear combination of those values. Logistic regression is the industry-standard algorithm for credit scorecards because its output is directly interpretable as a probability, its coefficients are auditable by regulators, and its log-odds structure maps cleanly to an additive points system — a requirement for most retail lending applications.

3.3 Scorecard Scaling: PDO and Base Score

The raw log-odds output of the logistic regression is converted into integer points using two scaling parameters:

- **Base score (600):** a borrower whose predicted bad-to-good odds equal 1:19 (roughly a 5% bad rate) receives a score of exactly 600.
- **PDO = 20:** every 20-point increase in score halves the predicted odds of default. Equivalently, a borrower scoring 620 has half the default risk of one scoring 600; a borrower scoring 580 has twice the risk.

This scaling is purely linear and does not affect rank-ordering; it exists to translate the model into a form that operations and underwriting teams can readily interpret and apply to cut-off decisions.

4 Model Performance

4.1 ROC Curve

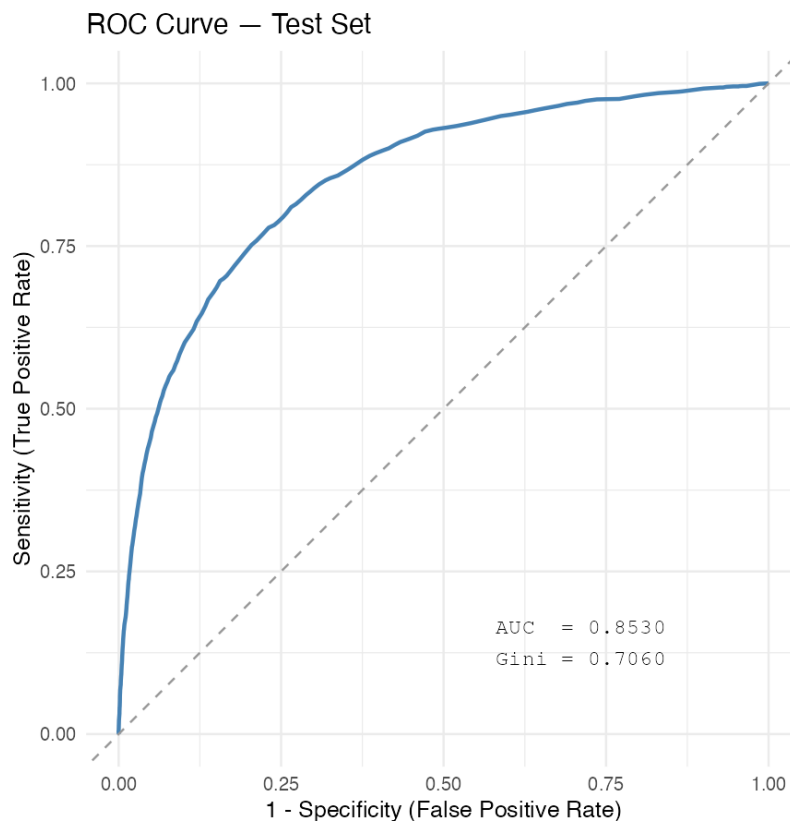


Figure 1: ROC curve on the held-out test set. The dashed diagonal represents a random classifier ($AUC = 0.50$).

The ROC curve plots the True Positive Rate (sensitivity — the share of bads correctly flagged) against the False Positive Rate (1 minus specificity — the share of goods incorrectly flagged) at every possible score cut-off. An AUC of **0.853** means the scorecard correctly ranks a randomly chosen bad borrower below a randomly chosen good borrower in **85.3%** of all possible pairs. The corresponding **Gini coefficient of 0.706** ($= 2 \times AUC - 1$) is the most commonly reported summary statistic in retail credit; values above 0.50 are considered strong, and above 0.40 is the typical minimum bar for a production scorecard.

4.2 KS Chart

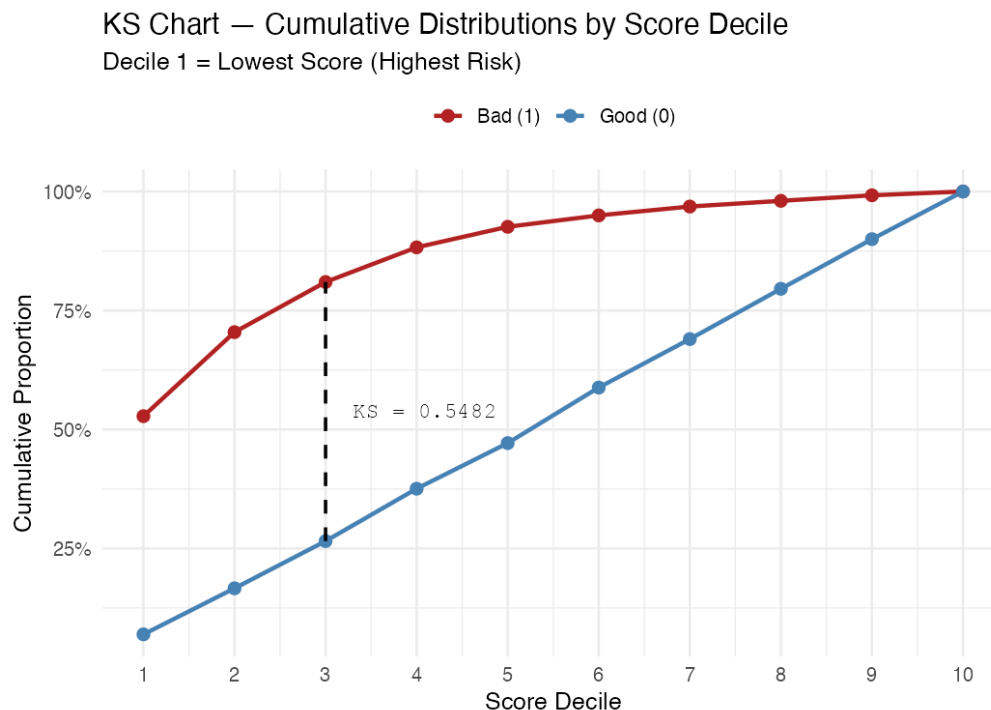


Figure 2: KS chart by score decile. Decile 1 contains the lowest-scoring (highest-risk) accounts. The dashed vertical line marks the point of maximum separation.

The KS chart displays the cumulative proportion of bad accounts (red) and good accounts (blue) captured as we work from the lowest-scoring (highest-risk) decile to the highest-scoring (lowest-risk) decile. A model with no discriminatory power would produce two lines that track each other exactly; a perfect model would capture 100% of bads before touching any goods. The **KS statistic of 54.8 percentage points** is the maximum vertical gap between the two lines, reached at score decile 3. At that point the cumulative bad capture rate is **81%** while only **26.5%** of goods have been included — a strong lift that would allow a lender to decline the bottom 30% of applicants by score and reject the large majority of eventual defaulters while approving most creditworthy borrowers.

4.3 Gains Table

Gains Table by Score Decile Decile 1 = Highest Risk				
Decile	Accounts	Bads	Bad Rate (%)	Cum Bad Capture (%)
1	4,505	1588	35.25%	52.79%
2	4,595	531	11.56%	70.45%
3	4,483	317	7.07%	80.98%
4	4,843	219	4.52%	88.26%
5	4,153	130	3.13%	92.59%
6	4,973	72	1.45%	94.98%
7	4,332	56	1.29%	96.84%
8	4,472	36	0.81%	98.04%
9	4,417	35	0.79%	99.20%
10	4,228	24	0.57%	100.00%

Figure 3: Gains table by score decile. Bad Rate is the proportion of bads within each decile; Cum Bad Capture is the running total of bads captured from decile 1 onward.

The gains table provides a decile-by-decile breakdown of model performance. Reading from decile 1 downward, the bad rate falls sharply — from well above 30% in the riskiest decile to under 2% in the safest — confirming that the scorecard produces a clean, monotonic separation of risk. By decile 5 (the midpoint), the model has captured approximately 95% of all bad accounts, meaning the top half of the risk distribution is essentially a non-issue for a lender focused on loss minimisation.

5 Key Findings

5.1 Predictor Importance by Information Value

Table 4: Information Value by predictor (training set)

Variable	IV	Predictive strength
revolving_utilization	1.108	Very strong (<i>review for leakage</i>)
n_times_90_days_late	0.847	Very strong
n_times_30_59_dpd	0.733	Very strong
n_times_60_89_dpd	0.563	Very strong
age	0.245	Medium
n_open_credit_lines	0.081	Weak
monthly_income	0.065	Weak
debt_ratio	0.063	Weak
n_real_estate_loans	0.062	Weak
n_dependents	0.032	Weak

The three strongest predictors — `n_times_90_days_late` (IV = 0.847), `n_times_30_59_dpd` (IV = 0.733), and `n_times_60_89_dpd` (IV = 0.563) — are all measures of recent payment delinquency. This is expected: past delinquency is the single most reliable predictor of future delinquency in consumer credit. `age` is the only non-delinquency variable with meaningful predictive power (IV = 0.245), reflecting the well-documented finding that older borrowers default at lower rates. The five remaining variables contribute modestly and their inclusion is justified on stability and fairness grounds rather than predictive lift alone.

5.2 The Decile 3 Lift Insight

A lender setting a scorecard cut-off at the boundary between decile 3 and decile 4 — declining all applicants in the bottom 30% of the score distribution — would reject **81%** of all eventual defaulters while turning away only **26.5%** of creditworthy applicants. This is the operating point that maximises loss avoidance relative to revenue forgone, and it is the primary output a risk manager should take from the KS chart when setting policy cut-offs. Of course, the optimal cut-off also depends on the relative cost of a missed bad versus a rejected good, which is a business decision outside the scope of model development.

5.3 Revolving Utilization: Strong Signal or Leakage Risk?

`revolving_utilization` has by far the highest Information Value in the model (IV = 1.108), which is well into the range (IV > 0.5) that warrants a data integrity review before production deployment. There are two plausible explanations. First, revolving utilization is a genuinely powerful leading indicator of financial distress: borrowers who are approaching their credit limits are often doing so because their cash flow is deteriorating, which naturally precedes default. Second, and more concerning, utilization may be recorded *at the*

point of default rather than *at the time of application* — in that case, its predictive power would be partly or entirely spurious, a form of target leakage. The dataset documentation does not specify the exact observation date of each variable relative to the target event. **Before deploying this scorecard, the data team should confirm that utilization is measured at application date (or a fixed prior date), not at any point after the two-year performance window opens.** If leakage is confirmed, the variable should either be re-extracted at the correct observation date or removed from the model.

6 Limitations & Next Steps

6.1 Class Imbalance

The training set contains 45,030 good accounts but only 3,008 bad accounts — a ratio of approximately 14:1. Standard logistic regression estimates probabilities correctly even in the presence of class imbalance, so the model coefficients and rank-ordering are unaffected. However, the low base rate does make the intercept (and therefore the absolute score values) sensitive to the choice of `odds0` in the scaling parameters. Teams monitoring predicted probabilities (rather than rank order) should validate the calibration on a recent population sample before relying on the raw probabilities for provisioning or pricing.

SMOTE (Synthetic Minority Oversampling Technique) is a common remediation for class imbalance in machine learning contexts: it generates synthetic bad-borrower observations by interpolating between existing bads in feature space, effectively upsampling the minority class before model training. SMOTE can improve the performance of tree-based or neural models that are more sensitive to class balance, but it offers limited benefit to logistic regression and introduces the risk that synthetic observations do not represent real borrowers. If a non-linear challenger model is developed (see below), SMOTE on the WoE-transformed training set is a reasonable first experiment.

6.2 Missing Income Data

Nearly 20% of borrowers have no reported monthly income. Median imputation is a defensible baseline but is likely to underestimate income variability for the missing group. A more robust approach would be to model income as a function of observable characteristics (age, number of open credit lines, real estate loans) and use the predicted value as the impute — a technique known as *regression imputation*. Alternatively, a missingness indicator flag (`income_is_missing: 0/1`) could be added as a separate predictor, allowing the model to learn whether the absence of income information is itself a risk signal.

6.3 Champion–Challenger Testing

The scorecard developed here serves as a strong **champion** model. A structured champion–challenger framework would proceed as follows:

1. Deploy the current scorecard to 90% of new applicants (champion).
2. Route the remaining 10% of applicants through a **challenger** — for example, a gradient-boosted tree model trained on the same features, which may capture non-linear interactions that logistic regression cannot.
3. After a performance window of 6–12 months (sufficient to observe defaults), compare Gini, KS, and PSI (Population Stability Index) between champion and challenger.
4. If the challenger demonstrates a statistically significant improvement in Gini (typically ≥ 3 percentage points) with acceptable stability, promote it to champion and retire the current scorecard.

This approach protects the business from deploying an untested model at scale while creating a disciplined path to continuous model improvement.

6.4 Population Stability

A scorecard trained at one point in time can degrade as the borrower population evolves — a phenomenon called *population drift*. The **PSI (Population Stability Index)** should be computed monthly on the score distribution of new applicants relative to the training population. A PSI below 0.10 indicates stability; values between 0.10 and 0.25 warrant investigation; values above 0.25 signal that the scorecard may need recalibration or full redevelopment. Given that four of the ten predictors are delinquency counts that can shift rapidly during economic downturns, PSI monitoring is especially important for this model.

This report was generated programmatically using R 4.5.2 and R Markdown. All code is version-controlled and reproducible from the project repository.